

MOHAMMED ABDUALLAH

AI Engineer | Computer Vision | Backend Development

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PROFESSIONAL SUMMARY

AI Engineer with 3+ years of experience designing, training, and deploying computer vision, machine learning, and backend systems in production. Skilled in real-time object detection and tracking (YOLO, TensorRT, ByteTrack, DeepSORT), deep learning frameworks (TensorFlow, PyTorch, Keras), and backend development with Python, Django, and FastAPI. Experienced in building LLM and Retrieval-Augmented Generation (RAG) applications, RESTful APIs, and scalable data pipelines, with a track record of optimizing models for edge deployment on embedded devices such as NVIDIA Jetson.

TECHNICAL SKILLS

Programming Languages: Python, C/C++, C#, Java, JavaScript

Computer Vision: OpenCV, YOLOv8, YOLO11, ByteTrack, DeepSORT, TensorRT, ONNX, RTSP, Image Processing, ROI Detection, Line-Crossing Detection

Machine Learning & Deep Learning: TensorFlow, PyTorch, Keras, Scikit-learn, CNNs, Model Training & Evaluation, Data Analysis

LLM & Generative AI: RAG, LangChain, Ollama, Weaviate, pgvector, Vector Databases, Prompt Engineering, AI Agents

Backend Development: Django, FastAPI, REST API Design, OAuth, JWT, Microservices Architecture, WebRTC, MQTT

Databases: PostgreSQL, MySQL, SQL Server, MongoDB, Redis

DevOps & Tools: Docker, Kubernetes, Git, Jenkins, GitLab CI, Linux, CI/CD

Embedded Systems: NVIDIA Jetson Nano, Raspberry Pi, Arduino

Testing: pytest, unittest

Soft Skills: Problem Solving, Critical Thinking, Team Collaboration, Communication, Time Management, Adaptability, Continuous Learning.

PROFESSIONAL EXPERIENCE

Backend Engineer — *Blue Elite Tech*

May 2024 – Apr 2026

- Designed and implemented scalable RESTful APIs enabling seamless communication between client applications and server-side systems.
- Engineered and optimized relational (PostgreSQL, MySQL, SQL Server) and NoSQL (MongoDB) databases, improving data consistency and query performance.
- Developed core backend business logic in Python using Django, supporting key application workflows.
- Integrated OAuth and JWT authentication protocols to secure backend services and protect sensitive data.
- Improved system scalability and response times through load balancing, Redis caching, and resource-usage optimization.
- Contributed to migrating a monolithic architecture to microservices using Docker and Kubernetes for containerization and orchestration.
- Built CI/CD pipelines with Jenkins and GitLab CI to automate testing and deployment of backend features.
- Authored technical documentation for APIs, database schemas, and backend processes, and reported progress to stakeholders.

Computer Vision Engineer — *RAM Group*

Feb 2023 – Mar 2024

- Developed and fine-tuned real-time object detection, tracking, and recognition algorithms for embedded platforms.
- Optimized computer vision models for resource-constrained embedded devices, reducing latency and improving throughput.
- Trained, deployed, and maintained machine learning models for object detection, classification, and segmentation.
- Collaborated with hardware engineers to integrate computer vision pipelines into embedded systems.
- Implemented rigorous QA and testing procedures to validate detection accuracy and system reliability.
- Maintained detailed documentation of algorithms, models, and integration processes for cross-functional stakeholders.

KEY PROJECTS

YOLO11 TensorRT Real-Time Object Detection

Technologies: YOLO11, TensorRT, OpenCV, Python

- Built a real-time object detection system accelerated with NVIDIA TensorRT for both live webcam and recorded video streams.
- Implemented adjustable confidence thresholds, FPS/detection-count overlays, and annotated video export.

Fashion RAG Chatbot

Technologies: RAG, Weaviate, FastAPI, Groq, Python

- Developed an e-commerce fashion chatbot using Retrieval-Augmented Generation with a Weaviate vector database.
- Implemented semantic search, BM25 keyword search, hybrid search, and metadata filtering, served through a FastAPI API.

MeetMesh — Self-Hosted Video Meeting Platform

Technologies: Django, Django Channels, Redis, WebRTC

- Built a self-hosted video conferencing application supporting real-time peer-to-peer audio/video and screen sharing.
- Used Django Channels and Redis to handle real-time signaling and meeting events at scale.

Real-Time Arabic Voice Assistant

Technologies: FastAPI, WebSocket, Gemini, Speech-to-Text, GPU Acceleration

- Built an end-to-end Arabic voice assistant transcribing speech, generating responses via Google Gemini, and replying with synthesized speech.
- Streamed audio over WebSocket to a FastAPI backend with GPU-accelerated speech processing for low-latency responses.

Customer Churn Classification

Technologies: TensorFlow, Keras, Scikit-learn, Python

- Built an end-to-end churn prediction system handling class imbalance, categorical encoding, and feature scaling.
- Saved the trained model and preprocessing pipeline for reuse in predictions on new customer data.

EDUCATION

Bachelor of Science in Informatics Engineering

Aleppo University — 2021

LANGUAGES

Arabic: Native

English: Professional Working Proficiency